

INTERNSHIP PROJECT • GROUP 2

Development of Smart Procurement & Purchase Order Management System

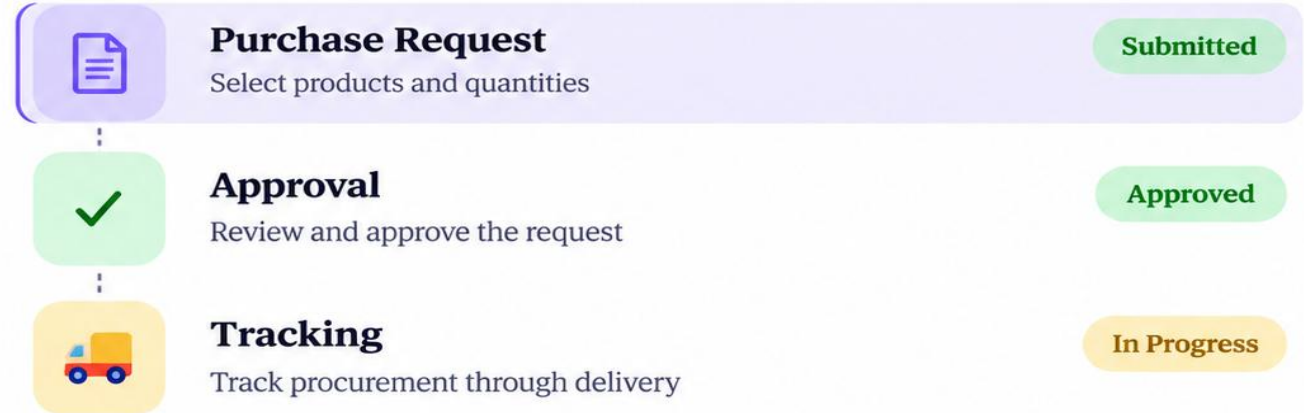
A structured digital workflow for requests, approvals, suppliers, payments, delivery tracking and feedback.

Gudipudi Durga Swathi

B.Tech • Infosys SpringBoard

PROCUREMENT WORKFLOW

From request to delivery.



HOW THIS STAGE WORKS

Create a purchase request

Employees select products, choose quantities and submit a request for approval.

Select products → Set quantity → Submit

Procurement progress

72%

Tracking
Order in progress

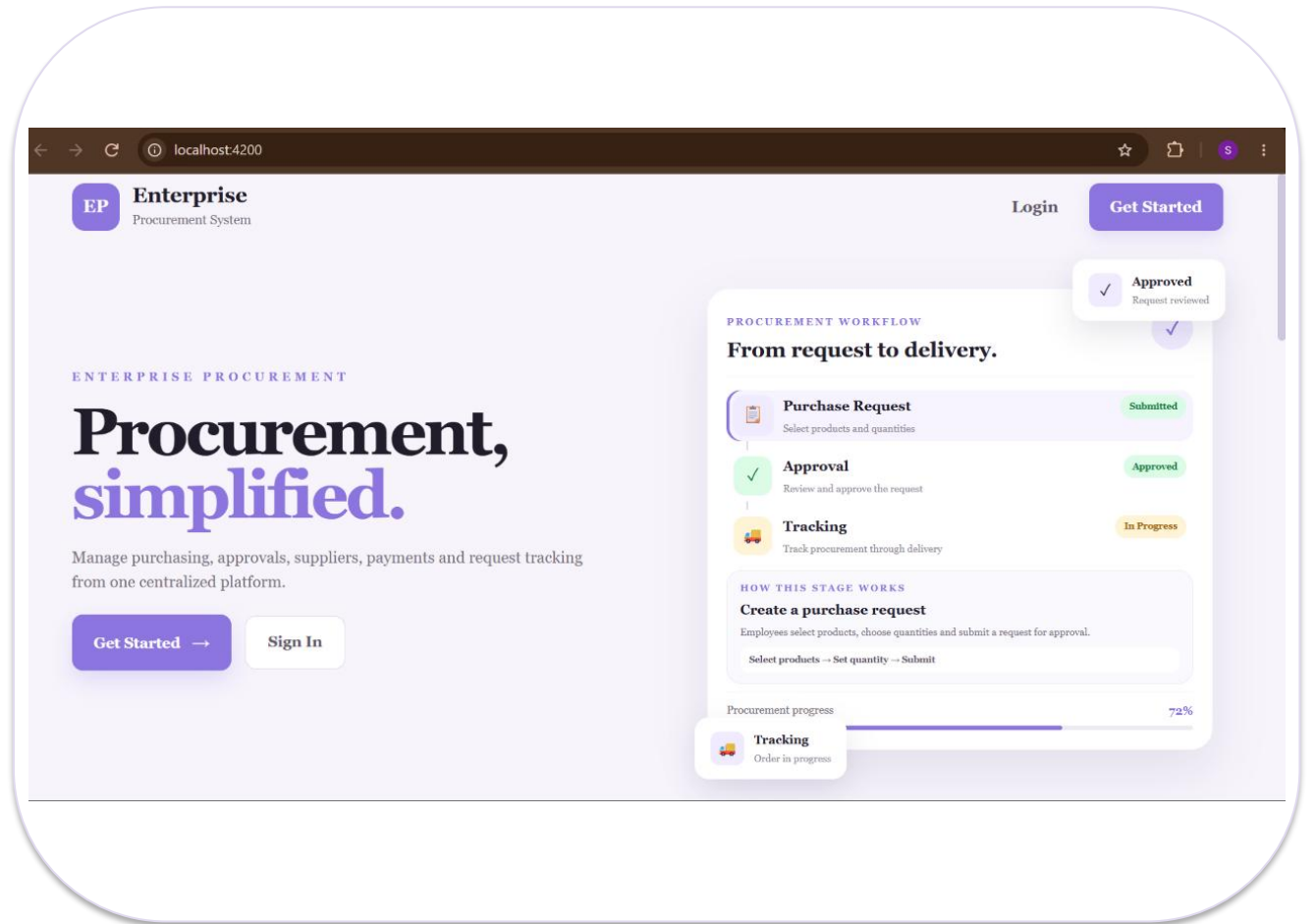
Project Introduction

What the system brings together in one procurement workflow.

THE IDEA

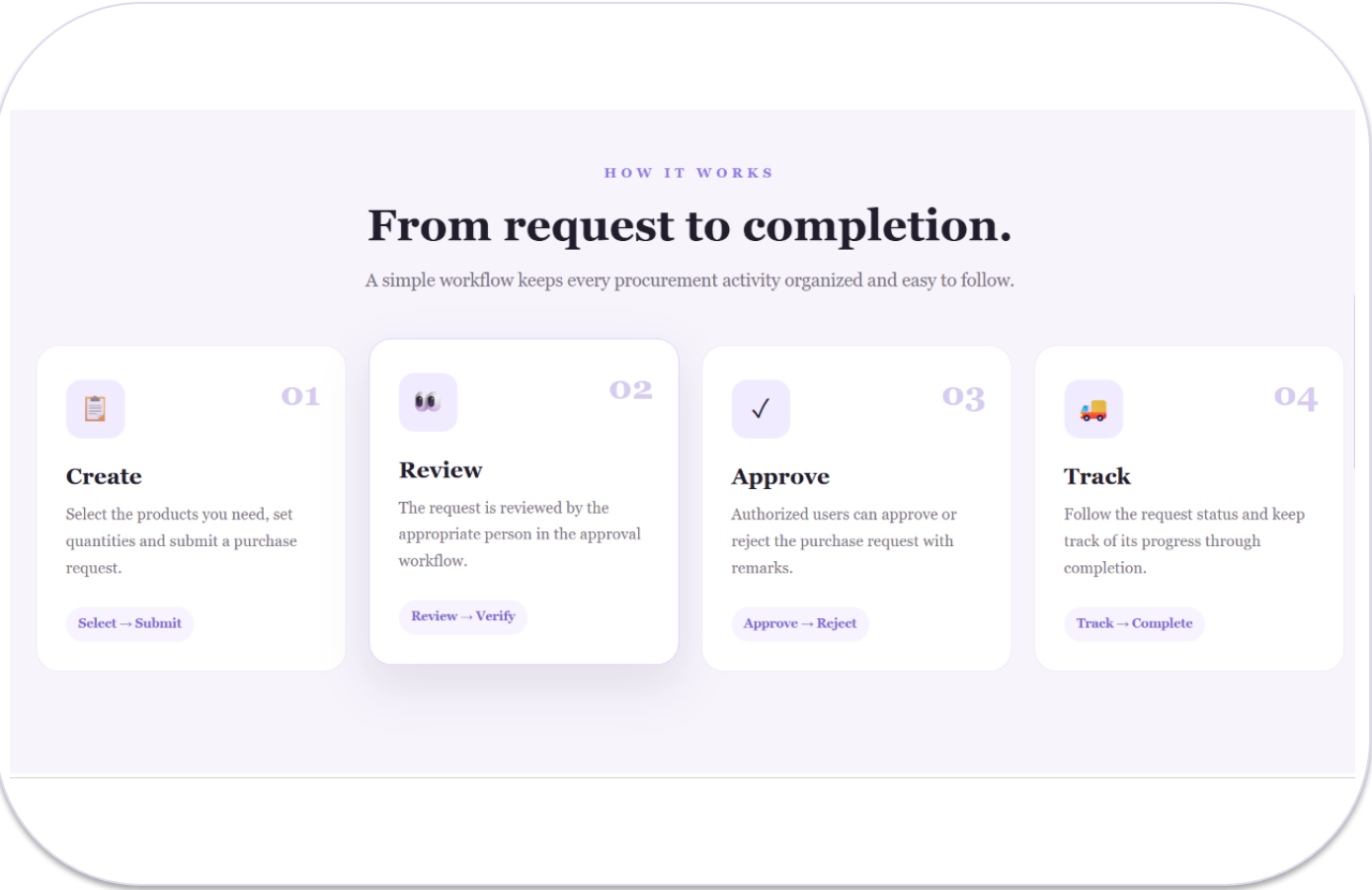
One workflow. Multiple roles. Clear status.

- Centralizes purchase requests and approvals.
- Connects products, suppliers, payments and delivery.
- Provides role-specific views for employees, admins and suppliers.
- Captures reports and feedback for visibility.



Objective & Need

From fragmented procurement activities to a traceable digital process.



WHY IT MATTERS

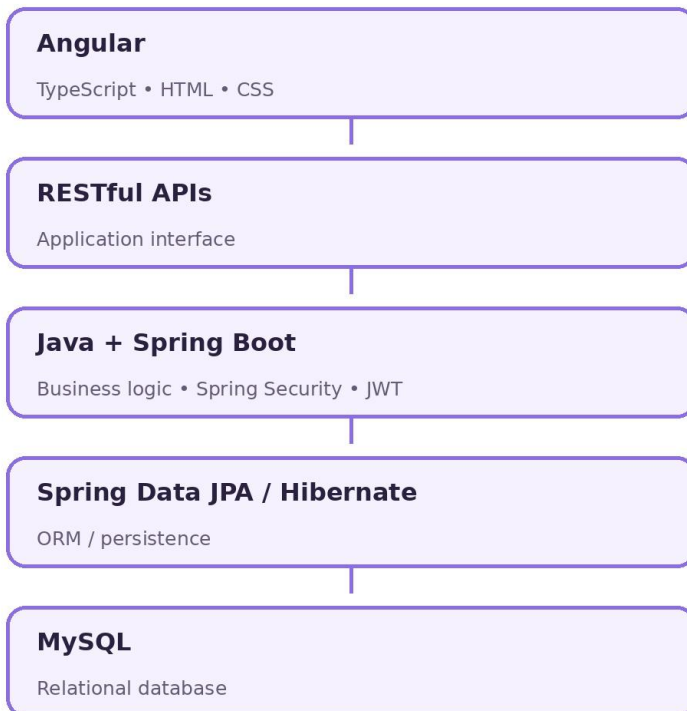
- Requests can be created and reviewed in a structured flow.
- Approval status is visible instead of relying on disconnected follow-ups.
- Supplier fulfilment, payment and delivery stages can be tracked.
- Dashboards and reports provide a consolidated operational view.

Goal: visibility across the full purchase lifecycle

Technology Stack & Architecture

Angular front end connected to Spring Boot services and MySQL persistence.

Technology Architecture



FRONTEND Angular • TypeScript • HTML • CSS

BACKEND Java • Spring Boot • RESTful APIs

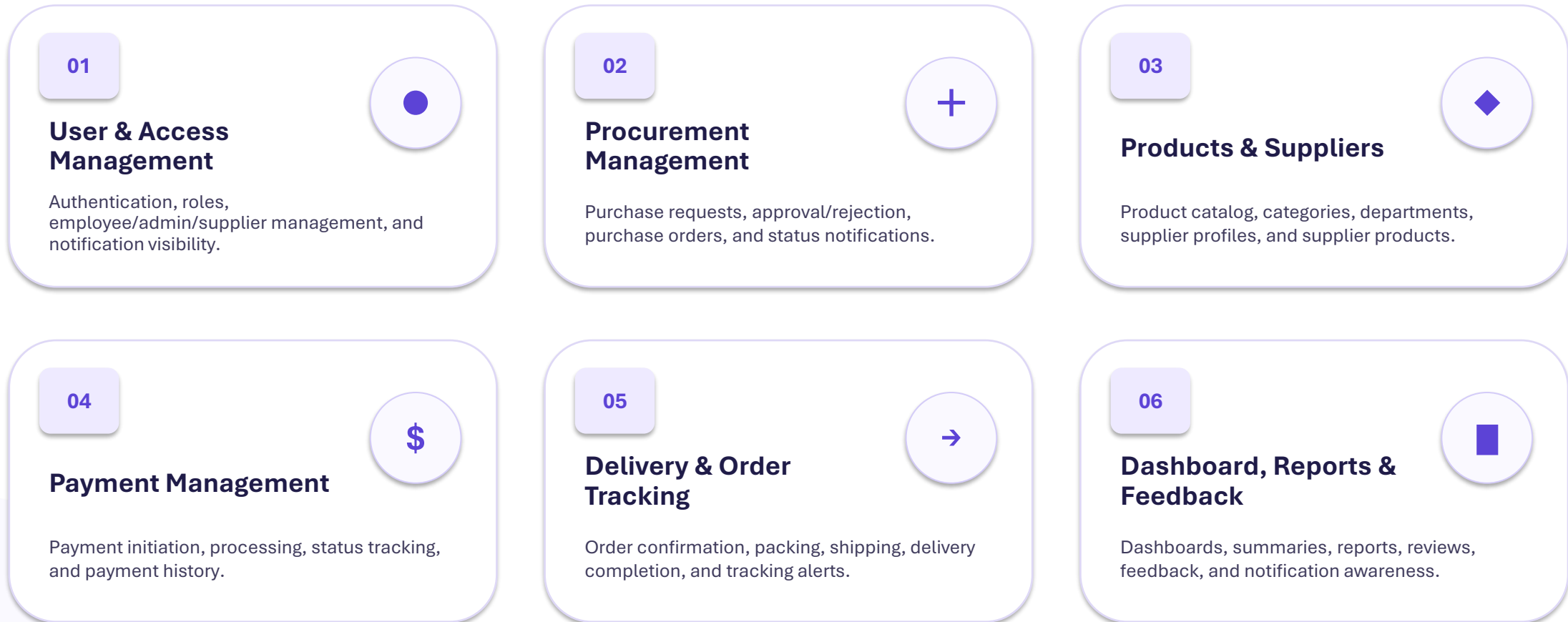
DATA Spring Data JPA / Hibernate • MySQL

SECURITY Spring Security • JWT

TOOLS Maven • IntelliJ IDEA • Postman • Git / GitHub

Six Functional Modules

The system is organized around six connected functional areas.



User & Access Management

Secure access, role-based control, and user administration.

01

Authentication

- Secure login with authentication
- Controlled entry to the procurement system
- Authentication for secure entry.

02

Role-Based Access

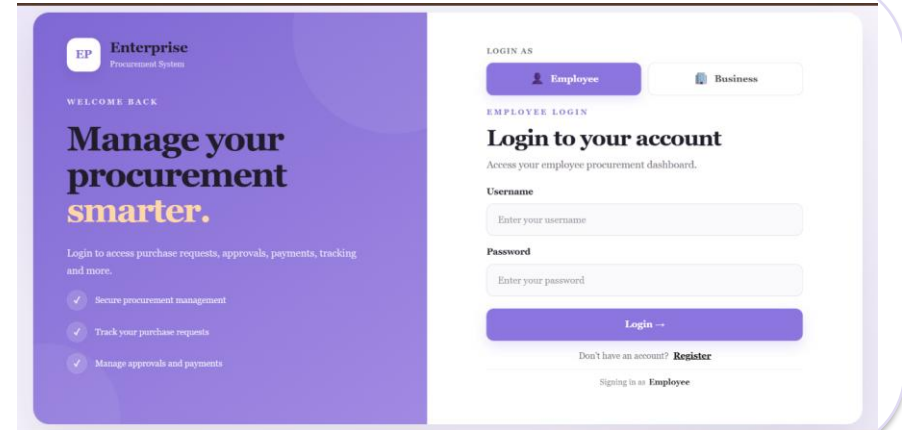
- Access based on assigned user role
- Employee, Admin, and Supplier permissions
- Role-based access for different users.

03

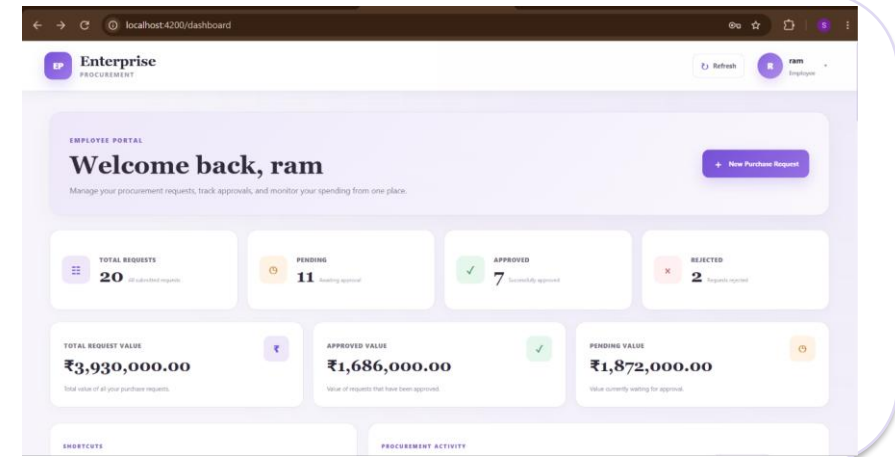
User Management

- Manage employee, admin, and supplier profiles
- Notification visibility for important updates

AUTHENTICATION • LOGIN



ROLE-BASED ACCESS • EMPLOYEE VIEW



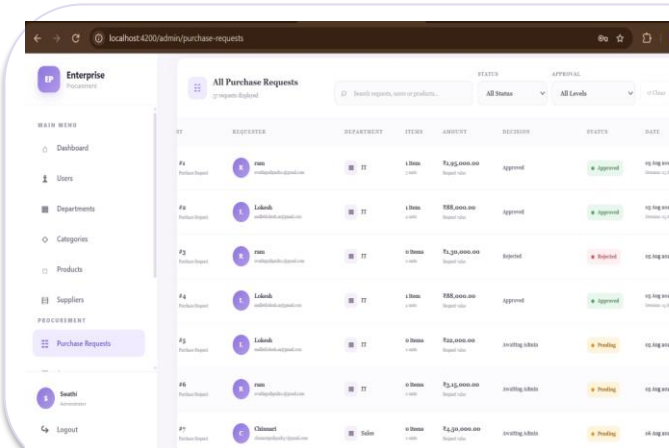
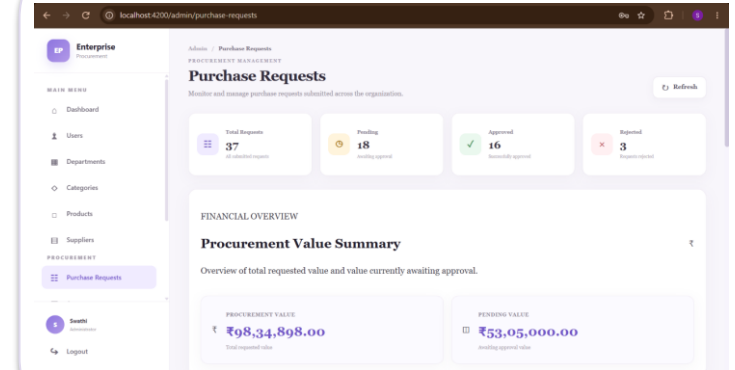
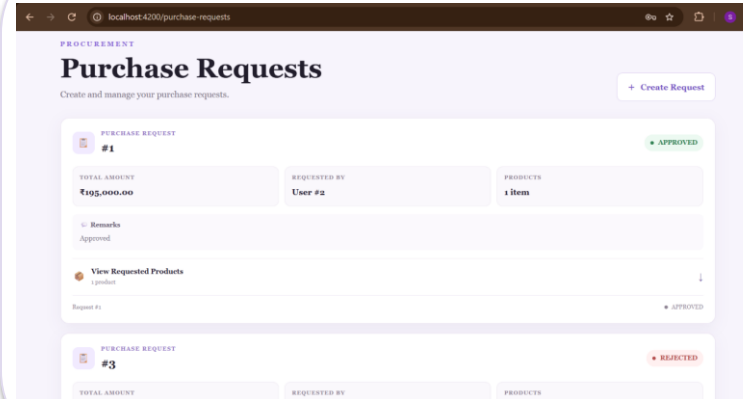
Procurement Management

A request moves through review and approval before becoming a purchase order.

PROCUREMENT FLOW

Request → Review → Approve

- Employees submit purchase requests.
- Admins review and approve or reject.
- Approved requests progress to purchase orders.
- Status is visible in the relevant dashboards.



Product & Supplier Management

Catalog and supplier information support procurement decisions and fulfilment.

Your business activity

Pending1Completed1Payment History2ReviewsProductsProfile

PURCHASE REQUEST

#33

APPROVED

REQUESTED BY

Laila

REQUEST DATE

08 Sept 2026

SUPPLIER AMOUNT

₹9,50,000.00

PAYMENT

Paid

Employee Remarks

Required for the staff

REQUESTED ITEMS

SAP S/4HANA Cloud Product #12

Qty: 1₹5,00,000.00

SAP Financial Management Product #13

Qty: 1₹4,50,000.00

DELIVERY TRACKING

Order #33

Packed

✓

Request Receipt

Purchase request received from the employee.

✓

Order Confirmed

Order confirmed after administrator approval.

✓

Order Packed

Products packed and ready for dispatch.

4

Order Shipped

Order has been dispatched to the employee.

5

Delivered

Order delivered to the employee.

NEXT STEP

Shipped

Mark as Shipped

WHAT IS MANAGED

PRODUCT CATALOG

Maintain product information for procurement.

CATEGORIES

Organize items into categories and departments.

SUPPLIER PROFILES

Keep supplier information available to the workflow.

SUPPLIER PRODUCTS

Associate products with supplier records.

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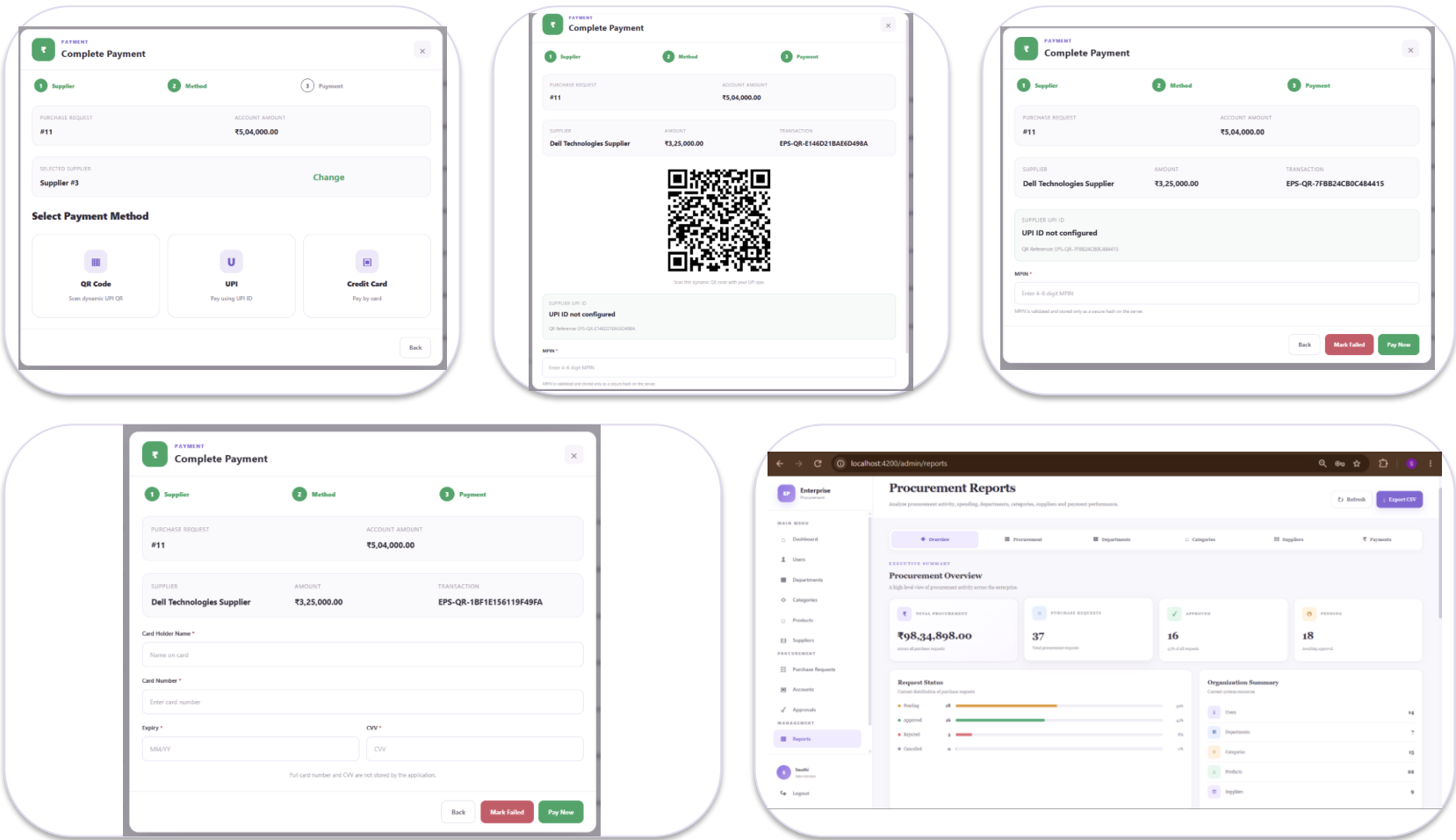
Payment Management

Payment screens show the progression from method selection to completed payment entry.

PAYMENT CAPABILITIES

- Initiate payment for procurement orders.
- Use available payment methods.
- Record payment status and history.
- Keep payment activity connected to the order workflow.

USER VIEWS



Delivery & Order Tracking

Track fulfilment stages from order confirmation through packing, shipping and completion.

TRACKING STAGES

1

Request Receipt

2

Order confirmation

3

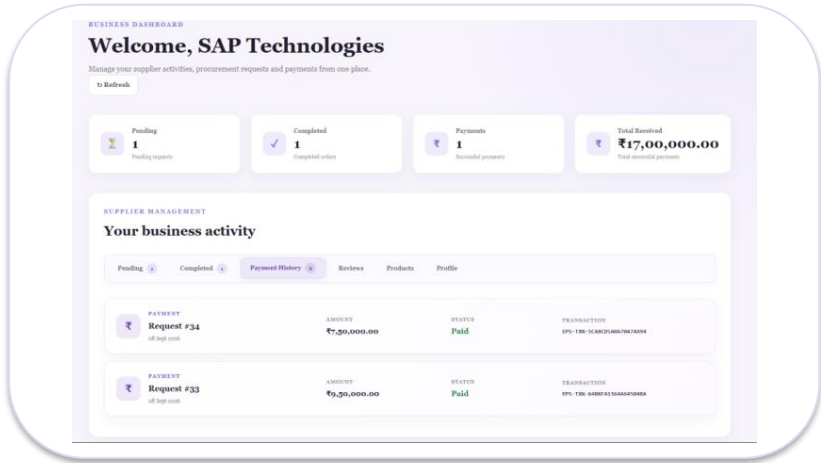
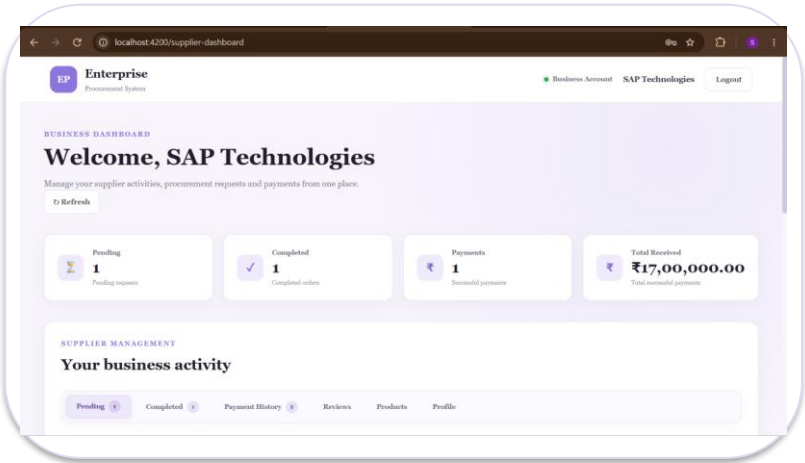
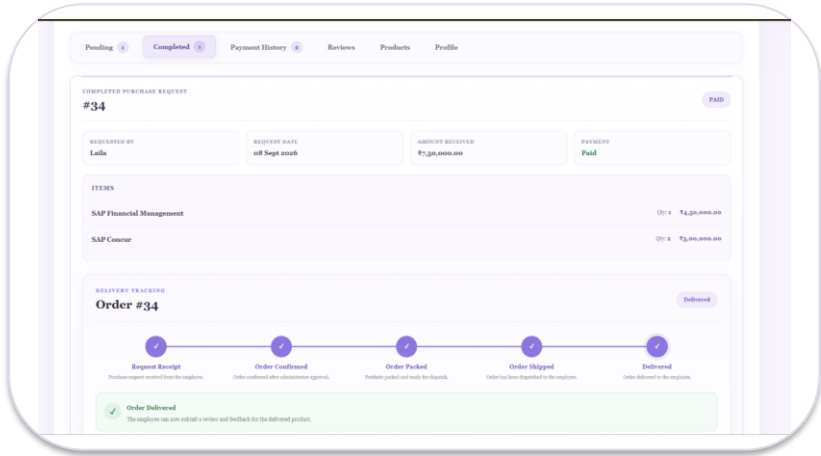
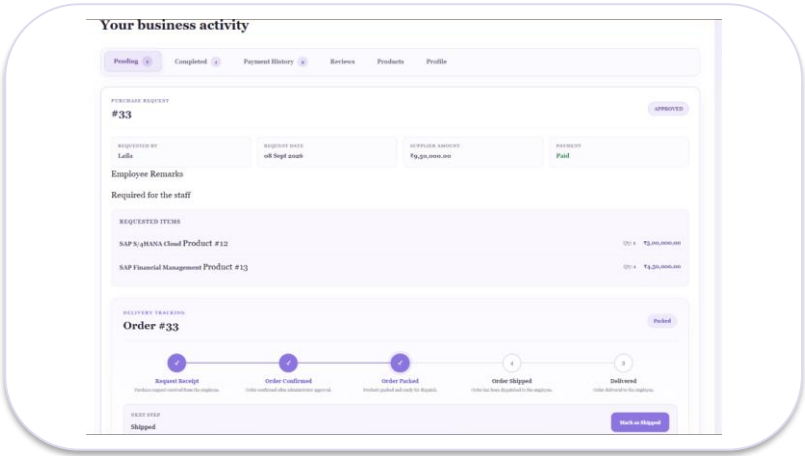
Packing

4

Shipping

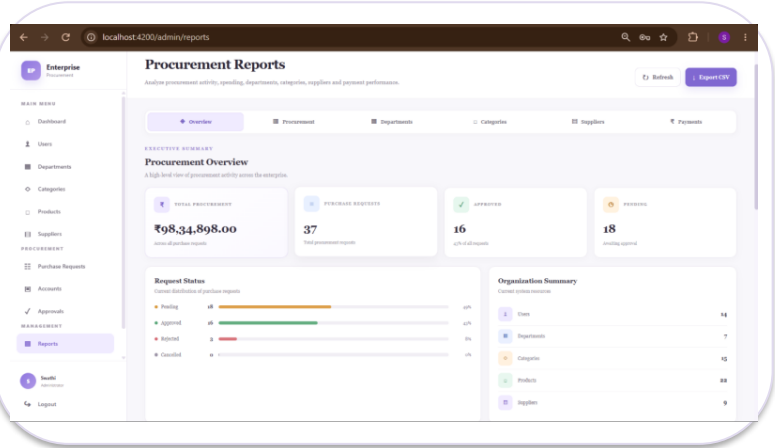
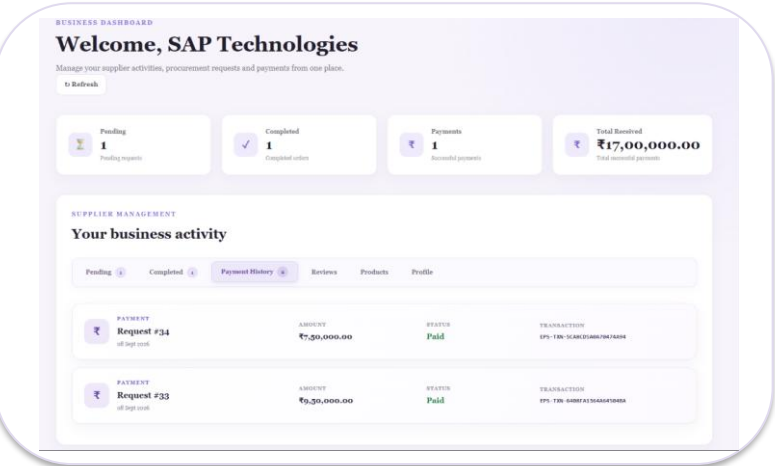
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Delivery completion



Dashboard, Reports & Feedback

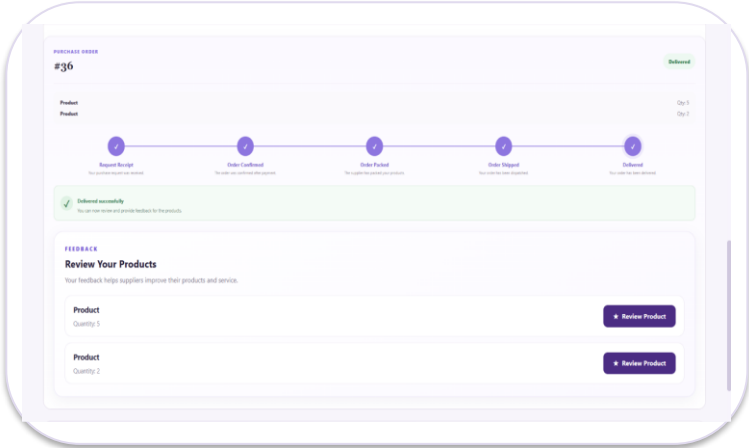
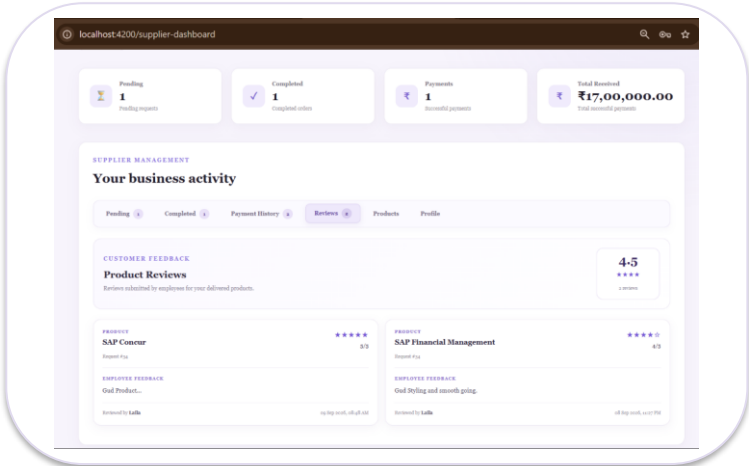
Operational visibility continues after procurement through reporting and user feedback.



DASHBOARDS & REPORTS

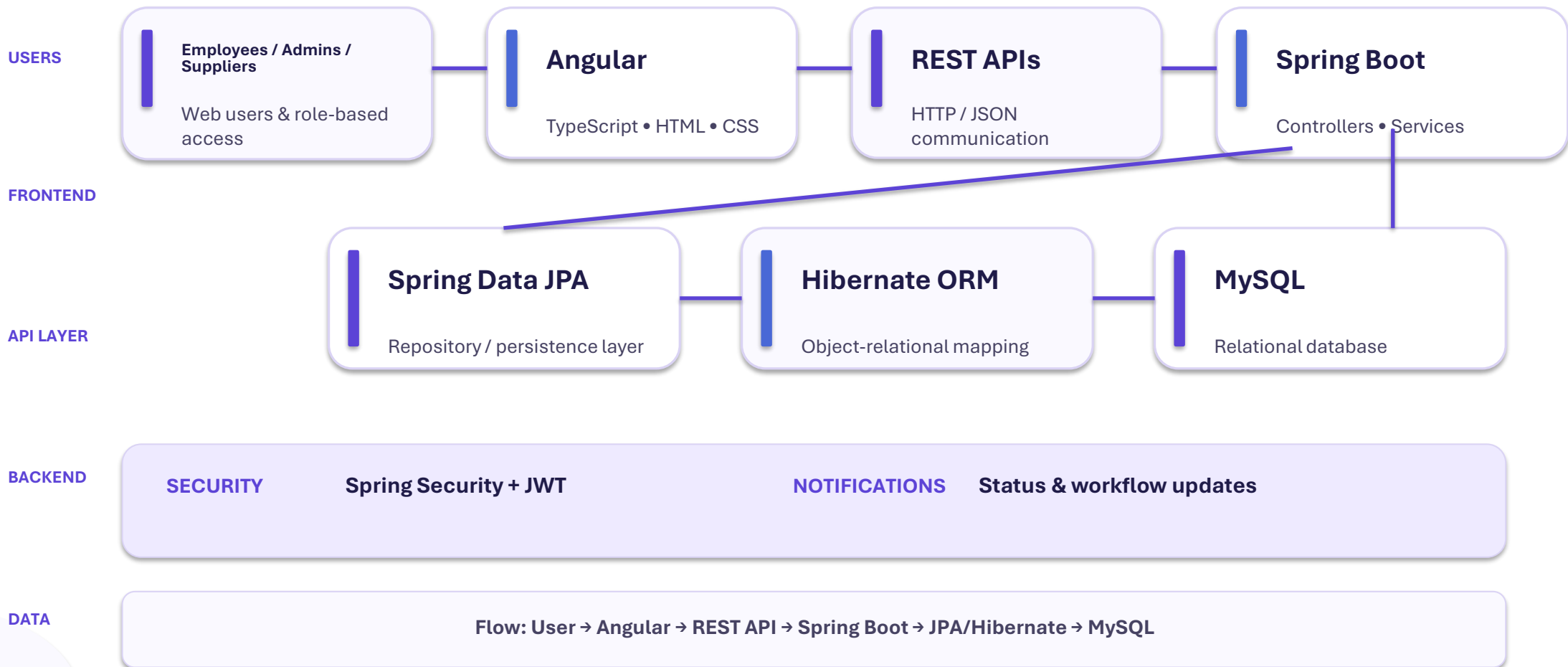
- Role-based dashboards surface relevant information.
- Procurement and payment summaries support visibility.
- Reports provide a consolidated view of activity.

Visibility → Review → Action



System Architecture

A layered architecture connecting users, Angular UI, REST APIs, Spring Boot services, and MySQL.



Thank You

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3

USER ROLES

6

FUNCTIONAL MODULES

4

DELIVERY STAGES

FROM REQUEST

DELIVERY → FEEDBACK

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